# DHAKA CITY OPEN GL PROJECT

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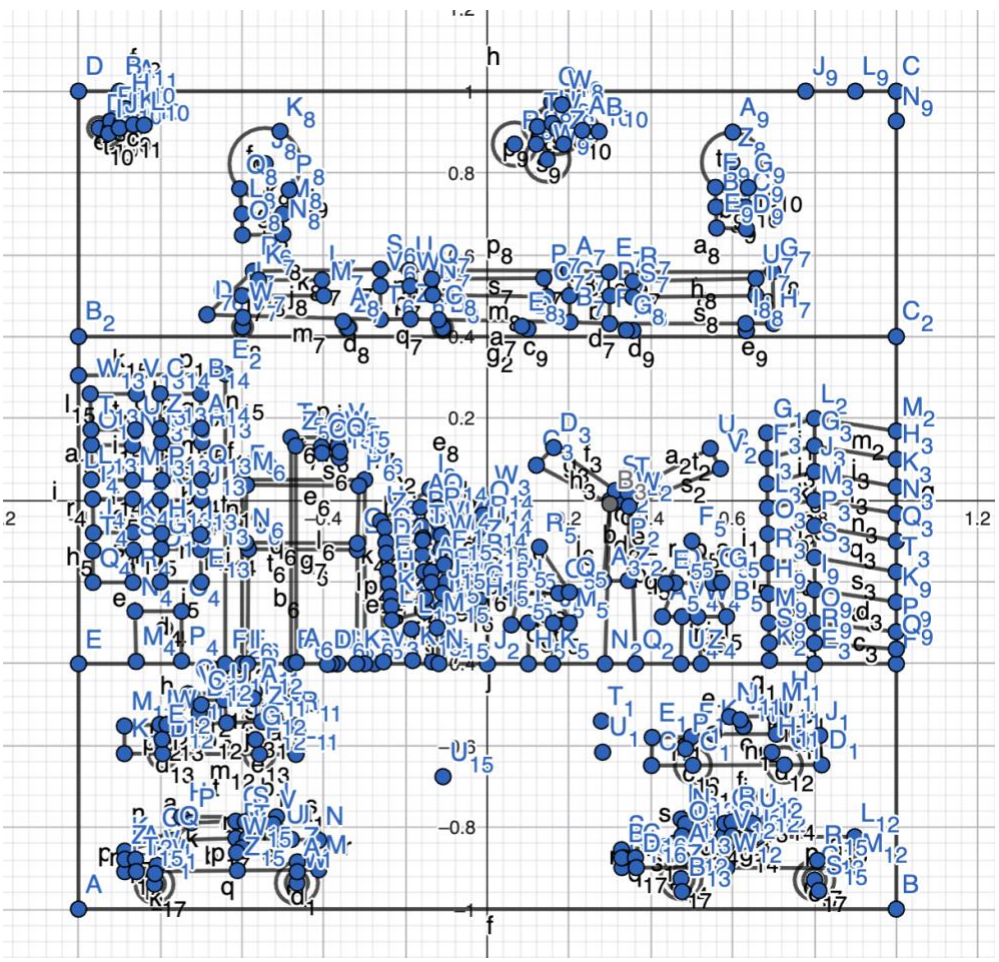
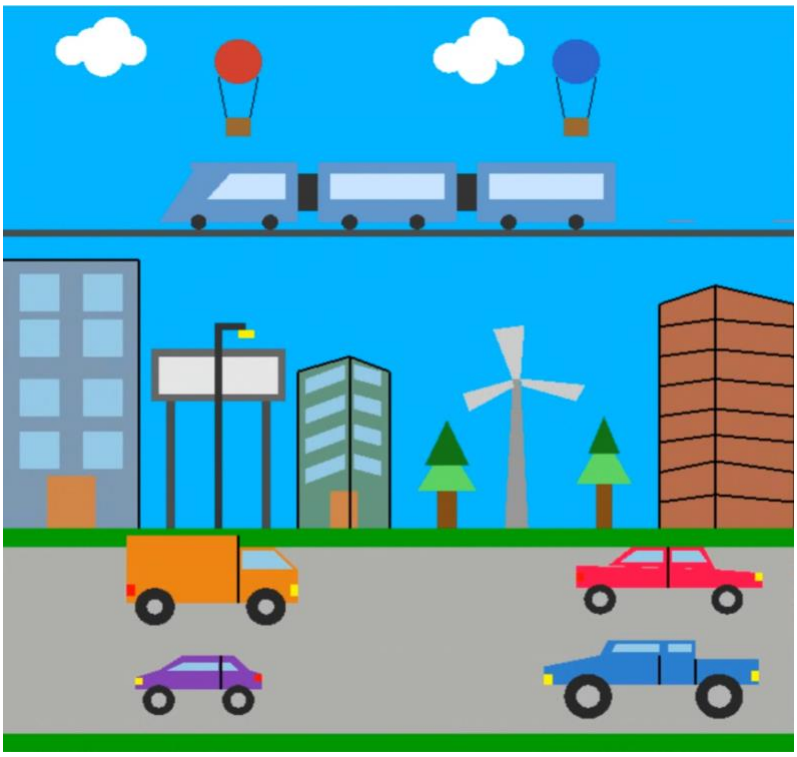
## Introduction

**Objects :** Cars, truck , road , windmill , trees , building ,billboard ,street light ,clouds ,hot air balloon , metro rail , metro line
**Animations:** cloud , windmill
**Mouse Interaction :** Metro rail start, metro rail stop
**Keyboard Interaction**: Upper lane cars start/stop , Lower lane cars start/stop , Hot Air baloons start/stop

This project shows a simple city scene with many objects like cars, trucks, roads, windmills, trees, buildings, billboards, street lights, clouds, hot air balloons, and a metro rail with its line. Some objects are animated, like moving clouds and a rotating windmill. You can control parts of the scene using inputs. With the mouse, you can start or stop the metro rail. With the keyboard, you can start or stop cars on both upper and lower roads, and also control the hot air balloons.

## *Project Graph





***List of objects assigning an object ID**

| SL# | Object Name | Object ID |
|---|---|---|
| 1 | Purple Car | carLD1 |
| 2 | Truck | carLU1 |
| 3 | blue car | carRD1 |
| 4 | red car | carRU1 |
| 5 | Road | road1 |
| 6 | Windmill | drawBase1 |
| 7 | Windmill Blades | drawBlades1 |
| 8 | Left Tree | treeLeft1 |
| 9 | Right Tree | treeRight1 |
| 10 | Right Building | buildingRight1 |
| 11 | Left Building | buildingLeft1 |
| 12 | Billboard | billboard1 |
| 13 | Street Light | streetLight1 |
| 14 | middle building | buildingR1 |
| 15 | Clouds | clouds1 |
| 16 | Hot Air Balloons | hotAirBalloons1 |
| 17 | Metro Rail | metroRail1 |
| 18 | Metro Line | metroline1 |

## *List of Functions To Represent Objects

| SL# | Function Name | Object_ID |
|---|---|---|
| 1 | carLD() | carLD1 |
| 2 | carLU() | carLU1 |
| 3 | carRD() | carRD1 |
| 4 | carRU() | carRU1 |
| 5 | road() | road1 |
| 6 | drawBase() | drawBase1 |
| 7 | drawBlades() | drawBlades1 |
| 8 | treeLeft() | treeLeft1 |
| 9 | treeRight() | treeRight1 |
| 10 | buildingRight() | buildingRight1 |
| 11 | buildingLeft() | buildingLeft1 |
| 12 | billboard() | billboard1 |
| 13 | streetLight() | streetLight1 |
| 14 | buildingR() | buildingR1 |
| 15 | clouds() | clouds1 |
| 16 | hotAirBalloons() | hotAirBalloons1 |
| 17 | metroRail() | metroRail1 |
| 18 | metroline() | metroline1 |

## *List of Animation Functions with ID

| SL# | Animation Function | Animation Function ID |
|---|---|---|
| 1 | display() | display1 |
| 2 | update() | update1 |
| 3 | handleKeypress() | Keypress |
| 4 | handleMouse() | Mouse |

## *Contribution

| Member Name | Implemented Functions | Implemented Animation Functions | Percentage of Contribution |
|---|---|---|---|
| Tamzid Islam | ALL | ALL | 100% |

## Conclusion

This OpenGL project presents a scenario of a typical day in Dhaka city. I have implemented the concepts I learned in computer graphics to build this project. I used translation to move objects by continuously updating their positions to create animation. I also applied rotation to animate the windmill blades and car wheels. In addition, I used keyboard and mouse inputs to make the system interactive. The code is organized into separate functions, making it easy to understand, modify, and maintain. This project helped me gain practical experience in animation, transformations, and user interaction. In the future, I can improve it further by adding better visuals, smoother animations, and more realistic effects.